1) a)

$$\lim_{x\to 2}\left(\frac{x^2-4}{x-2}\right)=\lim_{x\to 2}\left(\frac{2x}{1}\right)=4$$

1) b)

$$\lim_{x\to 1}\left(\frac{\ln x}{x-1}\right)=\lim_{x\to 1}\left(\frac{\frac{1}{x}}{1}\right)=1$$

1) c)

$$\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)=\lim_{x\to 0}\left(\frac{e^x}{1}\right)=1$$

1) d)

$$\lim_{x\to 0}\left(\frac{e^{2x}-1}{x}\right)=\lim_{x\to 0}\left(\frac{2e^{2x}}{1}\right)=2$$

1) e)

$$\lim_{x\to 0}\left(\frac{e^x-1}{\ln(1+x)}\right)=\lim_{x\to 0}\left(\frac{e^x}{\frac{1}{x+1}}\right)=1$$

1) f)

$$\lim_{x\to 0}\left(\frac{\ln(1+3x)}{x}\right)=\lim_{x\to 0}\left(\frac{\frac{3}{1+3x}}{1}\right)=3$$

2) a)

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{1}\right)=1$$

2) b)

$$\lim_{x\to 0}\left(\frac{1-\cos x}{x}\right)=\lim_{x\to 0}\left(\frac{\sin x}{1}\right)=0$$

2) c)

$$\lim_{x\to 0}\left(\frac{x}{\sin x}\right)=\lim_{x\to 0}\left(\frac{1}{\cos x}\right)=1$$

2) d)

$$\lim_{x\to 0}\left(\frac{x^2}{\sin x}\right)=\lim_{x\to 0}\left(\frac{2x}{\cos x}\right)=0$$

2) e)

$$\lim_{x\to 0}\left(\frac{-x+\tan x}{x}\right)=\lim_{x\to 0}\left(\frac{-1+\sec^2 x}{1}\right)=\lim_{x\to 0}\left(-1+\frac{1}{\cos^2 x}\right)=1$$

2) f)

$$\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)=\lim_{x\to 0}\left(\frac{3\cos 3x}{1}\right)=3$$

3) a)

$$\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)=\lim_{x\to 0}\left(\frac{\sin x}{2x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{2}\right)=\frac{1}{2}$$

3) b)

$$\lim_{x\to 0}\left(\frac{x-\sin x}{x^3}\right)=\lim_{x\to 0}\left(\frac{1-\cos x}{3x^2}\right)=\lim_{x\to 0}\left(\frac{\sin x}{6x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{6}\right)=\frac{1}{6}$$

3) c)

$$\lim_{x\to \infty}\left(\frac{\ln x}{x}\right)=\lim_{x\to \infty}\left(\frac{\frac{1}{x}}{1}\right)=\lim_{x\to \infty}\left(\frac{1}{x}\right)=0$$

3) d)

$$\lim_{x\to \infty}\left(\frac{x^2}{e^x}\right)=\lim_{x\to \infty}\left(\frac{2x}{e^x}\right)=\lim_{x\to \infty}\left(\frac{2}{e^x}\right)=0$$

3) e)

$$\lim_{x\to 0}\left(\frac{e^{3x}-1-x}{x^2}\right)=\lim_{x\to 0}\left(\frac{3e^{3x}-1}{2x}\right)=\lim_{x\to 0}\left(\frac{9e^{3x}}{2}\right)=\frac{1}{2}$$

3) f)

$$\lim_{x\to 0}\left(\frac{\ln(1+x)-x}{x^2}\right)=\lim_{x\to 0}\left(\frac{(x+1)^{-1}-1}{2x}\right)=\lim_{x\to 0}\left(\frac{-(x+1)^{-2}}{2}\right)=-\frac{1}{2}$$

3) g)

$$\lim_{x\to\infty}\left(\frac{\ln x}{\ln(x+1)}\right)=\lim_{x\to\infty}\left(\frac{\frac{1}{x}}{\frac{1}{x+1}}\right)=\lim_{x\to\infty}\left(\frac{x+1}{x}\right)=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)=1$$

4) a)

$$\lim_{x\to0}\left(\frac{\sinh x}{x}\right)=\lim_{x\to0}\left(\frac{\cosh x}{1}\right)=1$$

4) b)

$$\lim_{x\to0}\left(\frac{\tanh x}{2x}\right)=\lim_{x\to0}\left(\frac{\operatorname{sech}^2 x}{2}\right)=\frac{1}{2}$$

4) c)

$$\lim_{x\to0}\left(\frac{\sinh x-x}{4x^3}\right)=\lim_{x\to0}\left(\frac{\cosh x-1}{12x^2}\right)=\lim_{x\to0}\left(\frac{\sinh x}{24x}\right)=\lim_{x\to0}\left(\frac{\cosh x}{24}\right)=\frac{1}{24}$$

5) a)

$$\lim_{x\to0}\left(\frac{\arcsin x}{x}\right)=\lim_{x\to0}\left(\frac{\frac{1}{\sqrt{1-x^2}}}{1}\right)=1$$

5) b)

$$\lim_{x\to0}\left(\frac{\arctan x}{x}\right)=\lim_{x\to0}\left(\frac{\frac{1}{1+x^2}}{1}\right)=1$$

6) a)

They applied L'Hôpital's rule, but it wasn't in indeterminant form, because as $x\to0$, $\cos x+1\to2$ and $e^x+2\to3$

6) b)

$$\lim_{x\to0}\left(\frac{\cos x+1}{e^x+2}\right)=\frac{2}{3}$$

7)

$$\lim_{x\to\infty}(xe^{-2x})=\lim_{x\to\infty}\left(\frac{x}{e^{2x}}\right)=\lim_{x\to\infty}\left(\frac{1}{e^{2x}}\right)=0$$

8) a)

$$x\ln\left(1+\frac{1}{x}\right)=\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}$$

8) b)

$$\lim_{x\to\infty}\left(\frac{5\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}\right)=\lim_{x\to\infty}\left(\frac{\frac{-5x^{-2}}{1+\frac{1}{x}}}{-x^{-2}}\right)=\lim_{x\to\infty}\left(\frac{5}{1+\frac{1}{x}}\right)=5$$

9) a)

$$\lim_{x\to0}\left(\frac{xe^{x}}{\sin x}\right)=\lim_{x\to0}\left(\frac{xe^{x}+e^{x}}{\cos x}\right)=\frac{1}{1}=1$$

9) b)

$$\lim_{x\to0}\left(\frac{e^{x}\sin 7x}{\ln(4x+1)}\right)=\lim_{x\to0}\left(\frac{7e^{x}\cos 7x+e^{x}\sin 7x}{\frac{4}{4x+1}}\right)=\frac{7}{4}$$

9) c)

$$\lim_{x\to0}\left(\frac{\sin x-x\cos x}{x^{3}}\right)=\lim_{x\to0}\left(\frac{\cos x+x\sin x-\cos x}{3x^{2}}\right)=\lim_{x\to0}\left(\frac{x\sin x}{3x^{2}}\right)$$
$$=\lim_{x\to0}\left(\frac{x\cos x+\sin x}{6x}\right)=\lim_{x\to0}\left(\frac{-x\sin x+2\cos x}{6}\right)=\frac{1}{3}$$

9) d)

$$\lim_{x\to0}\left(\frac{1-\cos 5x+3x\sin 3x}{2x^{2}}\right)=\lim_{x\to0}\left(\frac{5\sin 5x+9x\cos 3x+3\sin 3x}{4x}\right)$$
$$=\lim_{x\to0}\left(\frac{25\cos 5x-27x\sin 3x+18\cos 3x}{4}\right)=\frac{43}{4}$$

10) a)

$$\lim_{x\to\pi}\left(\frac{\sin x}{x-\pi}\right)=\lim_{x\to\pi}\left(\frac{\cos x}{1}\right)=-1$$

10) b)

$$\lim_{x\to\pi}\left(\frac{x\cos x+\pi}{(x-\pi)^{2}}\right)=\lim_{x\to\pi}\left(\frac{-x\sin x+\cos x}{2(x-\pi)}\right)=\lim_{x\to\pi}\left(\frac{-x\cos x-2\sin x}{2}\right)=\frac{\pi}{2}$$

10) c)

$$\lim_{x\to\pi}\left(\frac{\cos\left(\frac{x}{2}\right)}{x\sin 2x}\right)=\lim_{x\to\pi}\left(\frac{-\frac{1}{2}\sin\left(\frac{x}{2}\right)}{2x\cos 2x+\sin 2x}\right)=\frac{-\frac{1}{2}}{2\pi}=-\frac{1}{4\pi}$$

11) a)

$$\lim_{a\to\infty}\left(\frac{a}{\mathrm{e}^{a}}\right)=\lim_{x\to\infty}\left(\frac{1}{\mathrm{e}^{a}}\right)=0$$

11) b)

Let $u=x$, $\frac{\mathrm{d}v}{\mathrm{d}x}=-\mathrm{e}^{-x}$ so $\frac{\mathrm{d}u}{\mathrm{d}x}=1$, $v=\mathrm{e}^{-x}$

So $\int x\mathrm{e}^{-x}\,\mathrm{d}x=x\mathrm{e}^{-x}+\int \mathrm{e}^{-x}\mathrm{d}x\ =x\mathrm{e}^{-x}-\frac{1}{\mathrm{e}^{x}}(+c)$

$$\therefore\int_0^{\infty}x\mathrm{e}^{-x}\,\mathrm{d}x=\lim_{b\to\infty}\int_0^{b}x\mathrm{e}^{-x}\,\mathrm{d}x=\lim_{b\to\infty}\left[x\mathrm{e}^{-x}-\frac{1}{\mathrm{e}^{x}}\right]_0^{b}=(0-0)-(0-1)=1$$

12) a)

$$\lim_{a\to\infty}\left(\frac{\ln a}{a}\right)=\lim_{a\to\infty}\left(\frac{\frac{1}{a}}{1}\right)=0$$

12) b)

Let $u=\ln x$, $\frac{\mathrm{d}v}{\mathrm{d}x}=x^{-2}$ so $\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x}$, $v=-x^{-1}$

So $\int\frac{\ln x}{x^{2}}\mathrm{d}x=-x^{-1}\ln x-\int\left(-x^{-1}\times\frac{1}{x}\right)\mathrm{d}x=-\frac{1}{x}\ln x+\int x^{-2}\,\mathrm{d}x$

$=-\frac{1}{x}\ln x-\frac{1}{x}(+c)$

$$\therefore\int_1^{\infty}\frac{\ln x}{x^{2}}\mathrm{d}x=\lim_{b\to\infty}\int_1^{b}\frac{\ln x}{x^{2}}\mathrm{d}x=\lim_{b\to\infty}\left[-\frac{1}{x}\ln x-\frac{1}{x}\right]_1^{b}=(0-0)-(-0-1)=1$$

13) a)

$$\frac{\mathrm{d}}{\mathrm{d}x}[\ln(\cos x)]=\frac{1}{\cos x}\times-\sin x=-\tan x$$

13) b)

$$\lim_{x\to0}\left(\frac{\ln(\cos x)}{x^{2}}\right)=\lim_{x\to0}\left(\frac{-\tan x}{2x}\right)=\lim_{x\to0}\left(\frac{-\sec^{2}x}{2}\right)=-\frac{1}{2}$$

14) a)

$\frac{\mathrm{d}}{\mathrm{d}x}[\ln(\cosh x)] = \frac{1}{\cosh x} \times \sinh x = \tanh x$

14) b)

$\lim_{x\to 0}\left(\frac{\ln(\cosh x)-7x^2}{x^2}\right) = \lim_{x\to 0}\left(\frac{\tanh x-14x}{2x}\right) = \lim_{x\to 0}\left(\frac{\operatorname{sech}^2 x-14}{2}\right) = \frac{1-14}{2} = -\frac{13}{2}$

15) a)

$\lim_{x\to 0^+}(x\ln x) = \lim_{x\to 0^+}\left(\frac{\ln x}{\frac{1}{x}}\right) = \lim_{x\to 0^+}\left(\frac{\frac{1}{x}}{-\frac{1}{x^2}}\right) = \lim_{x\to 0^+}(-x) = 0$

15) b)

$\ln x$ is only defined for $x > 0$ (in the real numbers). For $x < 0$, $\ln x$ is not real, so the expression $x\ln x$ is not defined. Therefore, we must take the limit as $x \to 0^+$ (from the right) only.

16)

$s(t) = \frac{\mathrm{e}^t-1}{t}$

$s'(t) = \frac{t\mathrm{e}^t-(\mathrm{e}^t-1)}{t^2} = \frac{t\mathrm{e}^t-\mathrm{e}^t+1}{t^2}$

$\lim_{t\to 0}\left(\frac{t\mathrm{e}^t-\mathrm{e}^t+1}{t^2}\right) = \lim_{t\to 0}\left(\frac{t\mathrm{e}^t+\mathrm{e}^t-\mathrm{e}^t}{2t}\right) = \lim_{t\to 0}\left(\frac{t\mathrm{e}^t}{2t}\right) = \lim_{t\to 0}\left(\frac{t\mathrm{e}^t+\mathrm{e}^t}{2}\right) = \frac{1}{2}$

$\therefore$ Velocity $= \frac{1}{2}$ ms^{-1}

17) a)

$\ln(1+t) = t - \frac{1}{2}t^2 + \frac{1}{3}t^3 - \frac{1}{4}t^4 + \cdots$

So $\ln(1+t) - t + \frac{1}{2}t^2 \approx t - \frac{1}{2}t^2 + \frac{1}{3}t^3 - t + \frac{1}{2}t^2 = \frac{1}{3}t^3$

17) b)

$\lim_{x\to 0}\left(x^{-3}\left(\ln(1+x) - x + \frac{1}{2}x^2\right)\right) \approx \lim_{x\to 0}\left(x^{-3}\left(\frac{1}{3}x^3\right)\right) = \frac{1}{3}$ as req.

17) c)

$\lim_{x\to 0}\left(\frac{\ln(1+x)-x+\frac{1}{2}x^2}{x^3}\right)=\lim_{x\to 0}\left(\frac{(1+x)^{-1}-1+x}{3x^2}\right)=\lim_{x\to 0}\left(\frac{-(1+x)^{-2}+1}{6x}\right)=\lim_{x\to 0}\left(\frac{2(1+x)^{-3}}{6}\right)=\frac{1}{3}$ as req.

17) d)

L'Hôpital's rule determines the value of the limit, whilst the Maclaurin expansion describes the behaviour near the point. In this particular case, the Maclaurin expansion identifies the dominant term to be $\frac{1}{3}x^3$, shows how the function approaches the limit, and provides a local approximation with error terms.

18)

Let $y=\left(1+\frac{1}{x}\right)^x$

So $\ln y=\ln\left[\left(1+\frac{1}{x}\right)^x\right]=x\ln\left(1+\frac{1}{x}\right)$

So $\lim_{x\to\infty}\ln y=\lim_{x\to\infty}\left(x\ln\left(1+\frac{1}{x}\right)\right)$

Rewrite in indeterminant form:

$\lim_{x\to\infty}\left(\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}\right)$

Apply L'Hôpital's rule:

$\lim_{x\to\infty}\left(\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}\right)=\lim_{x\to\infty}\left(\frac{\frac{-\frac{1}{x^2}}{1+\frac{1}{x}}}{-\frac{1}{x^2}}\right)=\lim_{x\to\infty}\left(\frac{1}{1+\frac{1}{x}}\right)=1$

So $\lim_{x\to\infty}(\ln y)=1$

Which means that $\ln y\to 1$ so that $y\to \mathrm{e}$

$\therefore \lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=\mathrm{e}$